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Subject Name: ADBMS

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Subject Code: 24CSP-310

QUESTION A:

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

Difference = Apples Sold - Oranges Sold

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

QUERY:

```
SELECT
    sale_date,
    SUM(
        CASE
            WHEN fruit = 'apples'
            THEN sold_num
            ELSE -sold_num
        END
    ) AS diff
FROM Sales
GROUP BY sale_date
ORDER BY sale_date;
```

RESULT:

	sale_date date 	diff bigint 
1	2020-05-01	2
2	2020-05-02	0
3	2020-05-03	20
4	2020-05-04	-1

QUESTION B:

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

- id represents the current node.
- p_id represents the parent node.

Classify every node into one of the following categories:

- Root: A node that does not have any parent.
- Leaf: A node that has a parent but does not have any children.
- Inner: A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

QUERY:

```
SELECT  
id,  
CASE  
WHEN p_id IS NULL THEN 'Root'  
WHEN id NOT IN (SELECT p_id FROM Tree WHERE p_id IS NOT NULL) THEN 'Leaf'  
ELSE 'Inner'  
END AS type  
FROM Tree;
```

RESULT:

	id [PK] integer	type text
1	1	Root
2	2	Inner
3	3	Leaf
4	4	Leaf
5	5	Leaf

